LAPORAN PRAKTIKUM

POSTTEST 1

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

Nama: Bintang Dhana Permadi (2509106014)

Kelas (A‘25)

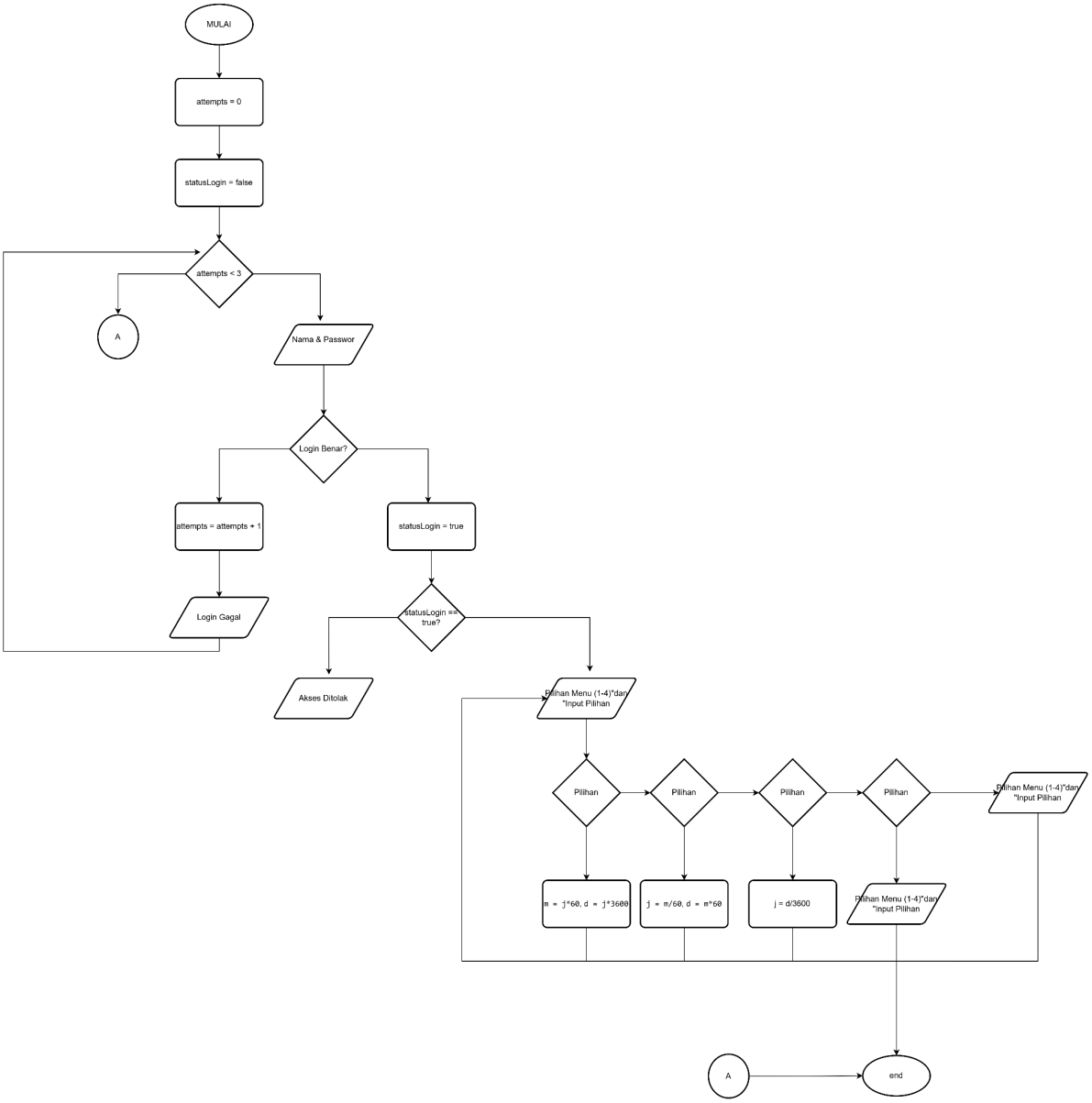
PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

SAMARINDA

2025

1. Flowchart



Gambar 1.1 Flowchart

2. Deskripsi Singkat Program

Program Konversi Waktu dengan Sistem Login

Program C++ ini berfungsi untuk mengonversi satuan waktu (Jam, Menit, Detik) secara interaktif. Program dilengkapi sistem keamanan **Login** dengan batas **3 kali percobaan**. Jika berhasil masuk, pengguna dapat melakukan berbagai konversi waktu berulang kali melalui menu utama hingga memilih untuk keluar. Program ini mengimplementasikan konsep dasar materi Modul 1 seperti variabel, operator aritmatika, percabangan (if-else), dan perulangan (while).

3. Source Code

Source Code:

```
#include <iostream>
#include <string>

using namespace std;

int main() {

    const string VALID_NAME = "bintang";
    const string VALID_PASS = "123";

    string nama, password;
    int attempts = 0;
    bool statusLogin = false;

    cout << "========================================\n";
    cout << "           SISTEM LOGIN                 \n";
    cout << "========================================\n";

    while (attempts < 3) {
        cout << "[User] Masukkan Nama       : ";
        cin >> nama;
        cout << "[Kunci] Masukkan Password  : ";
        cin >> password;
```

```
        if (nama == VALID_NAME && password == VALID_PASS) {
            cout << "\n[+] Login Berhasil! Selamat datang, " <<
nama << ".\n";
            statusLogin = true;
            break;
        } else {
            attempts++;
            cout << "[-] Login Gagal! (Sisa percobaan: " << 3 -
attempts << ")\n\n";
        }
    }

    if (statusLogin) {
        string pilihan;
        while (true) {
            cout <<
"\n========================================\n";
            cout << "          MENU KONVERSI WAKTU           \n";
            cout << "========================================\n";
            cout << " [1] Jam   --> Menit & Detik\n";
            cout << " [2] Menit --> Jam & Detik\n";
            cout << " [3] Detik --> Jam & Menit\n";
            cout << " [4] Keluar\n";
            cout << "----------------------------------------\n";
            cout << "Pilih menu (1/2/3/4): ";
            cin >> pilihan;

            if (pilihan == "1") {
                double jam;
                cout << "\n--- Konversi Jam ---\n";
                cout << "Masukkan jumlah Jam: ";
                cin >> jam;
                cout << "=> Hasil: " << jam << " Jam = " << jam *
60 << " Menit dan " << jam * 3600 << " Detik\n";

            } else if (pilihan == "2") {
                double menit;
                cout << "\n--- Konversi Menit ---\n";
                cout << "Masukkan jumlah Menit: ";
                cin >> menit;

                cout << "=> Hasil: " << menit << " Menit = " <<
```

```
menit / 60.0 << " Jam dan " << menit * 60.0 << " Detik\n";

            } else if (pilihan == "3") {
                double detik;
                cout << "\n--- Konversi Detik ---\n";
                cout << "Masukkan jumlah Detik: ";
                cin >> detik;

                cout << "=> Hasil: " << detik << " Detik = " <<
detik / 3600.0 << " Jam dan " << detik / 60.0 << " Menit\n";

            } else if (pilihan == "4") {
                cout << "\nTerima kasih telah menggunakan program
ini. Sampai jumpa!\n";
                break;
            } else {
                cout << "\n[!] Pilihan tidak valid!\n";
            }
        }
    } else {
        cout << "[!] Anda telah gagal 3 kali. Program
berhenti.\n";
    }

    return 0;
}
```

Gambar 3.1 sourcecode

4. Hasil Output

```
==========================================
            SISTEM LOGIN
==========================================
[User] Masukkan Nama        : bintang
[Kunci] Masukkan Password   : 123
```

Gambar 4.1 output menu login

```
[+] Login Berhasil! Selamat datang, bintang.

==========================================
           MENU KONVERSI WAKTU
==========================================
 [1] Jam   --> Menit & Detik
 [2] Menit --> Jam & Detik
 [3] Detik --> Jam & Menit
 [4] Keluar
------------------------------------------
Pilih menu (1/2/3/4):
```

Gambar 4.2 output menu konversi waktu/pilihan

```
==========================================
          MENU KONVERSI WAKTU
==========================================
 [1] Jam   --> Menit & Detik
 [2] Menit --> Jam & Detik
 [3] Detik --> Jam & Menit
 [4] Keluar
------------------------------------------
Pilih menu (1/2/3/4): 2

--- Konversi Menit ---
Masukkan jumlah Menit: 2
=> Hasil: 2 Menit = 0.0333333 Jam dan 120 Detik
```

Gambar 4.3 output menu ketika memilih konversi waktu/pilihan

```
========================================
          MENU KONVERSI WAKTU
========================================
 [1] Jam   --> Menit & Detik
 [2] Menit --> Jam & Detik
 [3] Detik --> Jam & Menit
 [4] Keluar
----------------------------------------
Pilih menu (1/2/3/4): 4

Terima kasih telah menggunakan program ini. Sampai jumpa!
PS D:\kuliah\SEMESTER-2\praktikum-apl\post-test\post-test-apl-1>
```

Gambar 4.4 output menu ketika selesai

5. Langkah-langkah GIT

5.1 GIT Init

```
Terima kasih telah menggunakan program ini. Sampai jumpa!
PS D:\kuliah\SEMESTER-2\praktikum-apl\post-test\post-test-apl-1> git init
Reinitialized existing Git repository in D:/kuliah/SEMESTER-2/praktikum-apl/post-test/post-test-apl-1
/.git/
```

Gambar 5.1 git init

5.2 GIT Add

```
PS D:\kuliah\SEMESTER-2\praktikum-apl\post-test\post-test-apl-1> git add .
```

Gambar 5.2 git add

5.3 GIT Commit

```
PS D:\kuliah\SEMESTER-2\praktikum-apl\post-test\post-test-apl-1> git commit -m "postest1"
[master aa5fbdd] postest1
 1 file changed, 0 insertions(+), 0 deletions(-)
```

Gambar 5.3 git commit

5.4 GIT Push

```
PS D:\kuliah\SEMESTER-2\praktikum-apl\post-test\post-test-apl-1> git push -u origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 296 bytes | 296.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Bintang-2727/praktikum-apl.git
   ab4d387..aa5fbdd  master -> master
branch 'master' set up to track 'origin/master'.
PS D:\kuliah\SEMESTER-2\praktikum-apl\post-test\post-test-apl-1>
```

Gambar 5.4 Push